Subramani B M_PES1UG24CS473_BPS#42

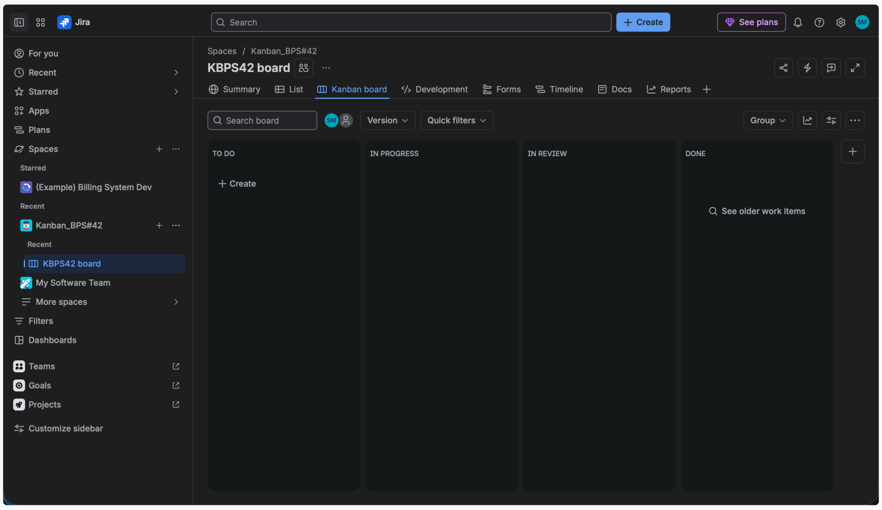
Lab 2: Jira Kanban Project Implementation Report

Course: Software Engineering Lab 2 • Problem Statement #42: Internal Microservice Catalog & Health Portal

1. Kanban Project & Board Creation

Project: Kanban_BPS#42 | Key: KBPS42

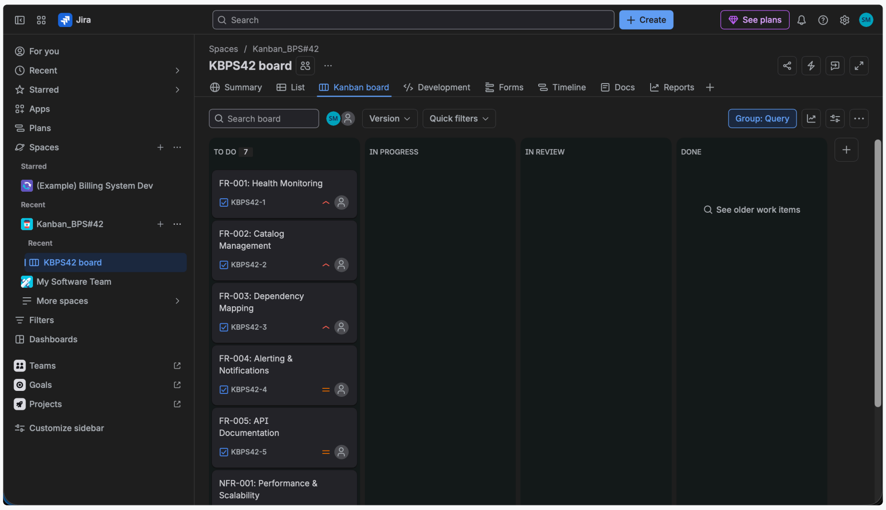
Description: Successfully initialized a new Company-Managed Kanban Software Development space in Jira named Kanban_BPS#42 with autogenerated project key KBPS42. The board contains the standard workflow stages: *TO DO*, *IN PROGRESS*, *IN REVIEW*, and *DONE*.



2. Requirements Addition (5 FRs + 2 NFRs)

7 Requirements Modeled

Description: Added all 5 Functional Requirements (FR–001 to FR–005) and 2 Non-Functional Requirements (NFR–001, NFR–002) as discrete work items on the board. Configured appropriate priority levels (High / Medium) based on stakeholder impact.



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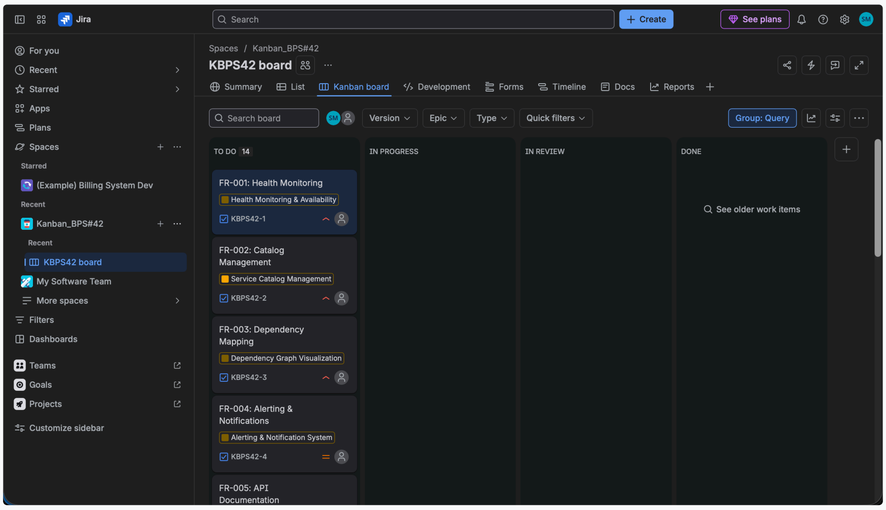
3. Epic Creation & Task Association

Epics Linked via Parent Relationships

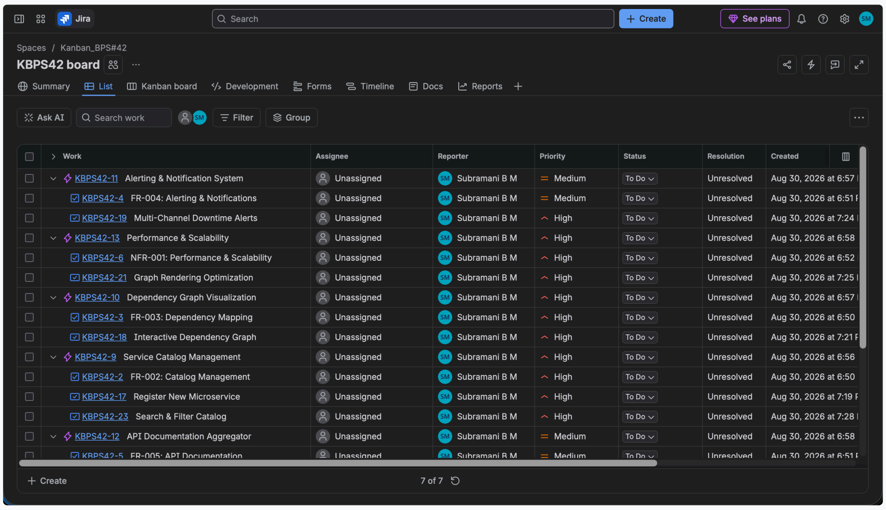
Description: Defined 7 comprehensive Epics corresponding to the core architectural domains. Assigned each requirement task to its respective parent Epic (e.g., *Health Monitoring & Availability*, *Service Catalog Management*, *Dependency Graph Visualization*), visible via colored Epic badges on the Kanban cards.



4. User Stories for Use Cases

List View: Epics & Nested User Stories

Description: Created user stories formatted as *"As a [role], I want [goal], So that [benefit]"* mapped under each Epic (e.g., `KBPS42-19: Multi-Channel Downtime Alerts`, `KBPS42-21: Graph Rendering Optimization`, `KBPS42-17: Register New Microservice`) with reporter `Subramani B M`.



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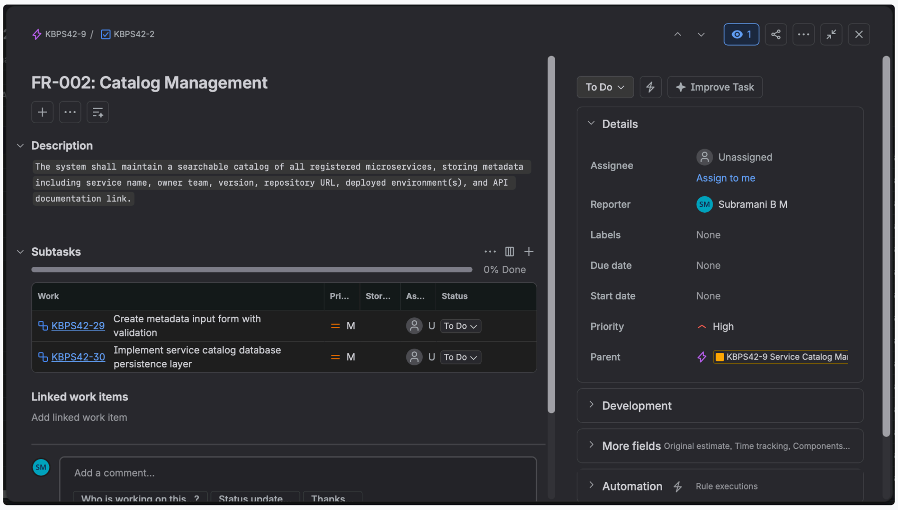
5. Task & Subtask Breakdown

Detailed Work Breakdown (FR-002 / KBPS42-2)

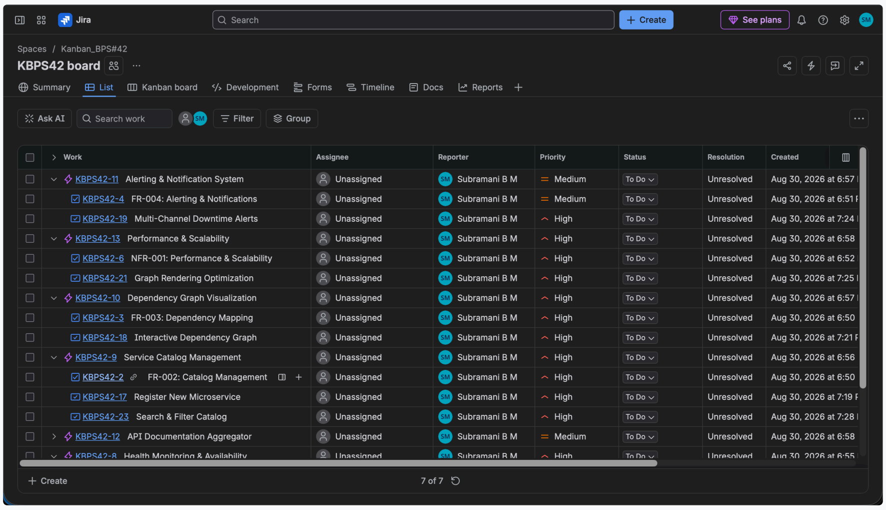
Description: Detailed engineering work breakdown demonstrating subtask creation under `FR-002: Catalog Management`: `KBPS42-29: Create metadata input form with validation` and `KBPS42-30: Implement service catalog database persistence layer`.



6. Comprehensive Backlog List View

Complete Backlog Hierarchy

Description: Complete hierarchical view of the project backlog showing all 7 Epics expanded with their constituent functional requirements, user stories, priorities, and assigned workflows ready for agile delivery.



Scrum_Project-Subramani B M_PES1UG24CS473-BPS#42

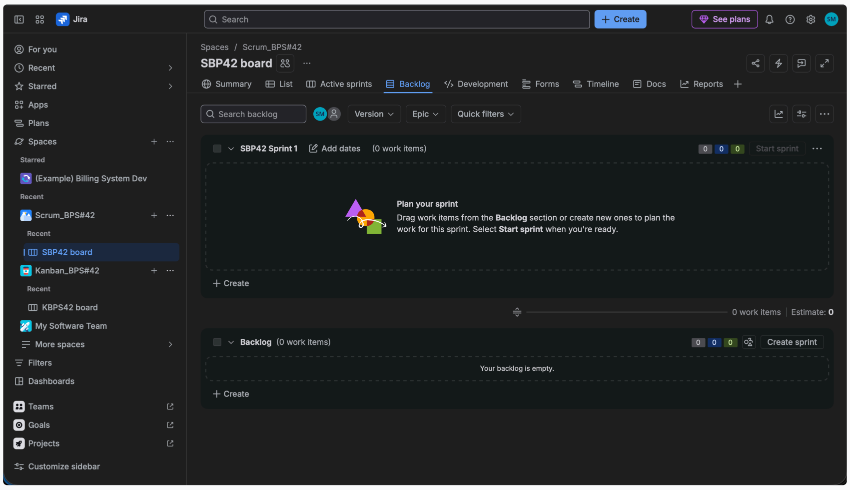
Lab 2: Jira Scrum Project & Sprint Simulation Report

Course: Software Engineering Lab 2 • Problem Statement #42: Internal Microservice Catalog & Health Portal

1. Scrum Project & Sprint Planner Setup

Project: Scrum_BPS#42 | Key: SBP42

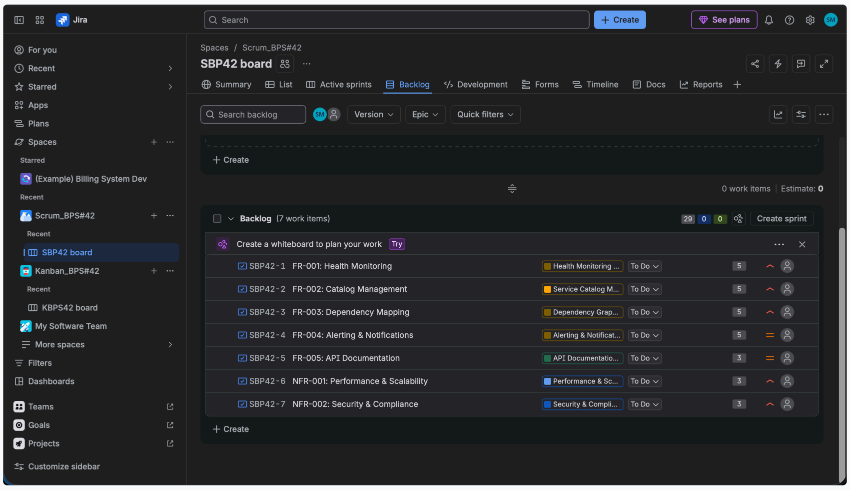
Description: Initialized a Company-Managed Scrum Software Development project named `Scrum_BPS#42` with project key SBP42, setting up the agile backlog and sprint planning workspace.



2. Product Backlog with Story Points & Epics

Total Story Points: 29 (Fibonacci Estimation)

Description: Populated the product backlog with all 7 requirements (5 FRs + 2 NFRs) mapped to their parent Epics. Assigned relative effort estimates using the Fibonacci scale (5 pts for high complexity, 3 pts for supporting features) totaling 29 story points.



Lab 2: Jira Scrum Project & Sprint Simulation Report

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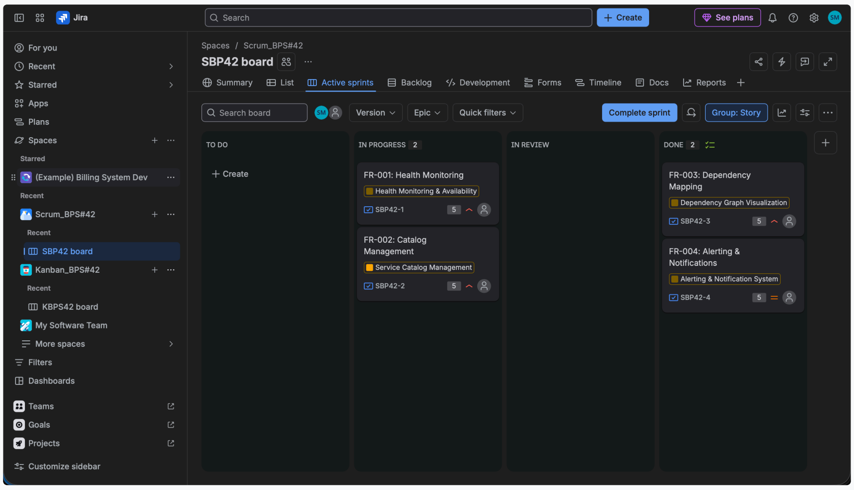
3. Active Sprint Board (Sprint 1 Simulation)

1-Week Sprint Duration

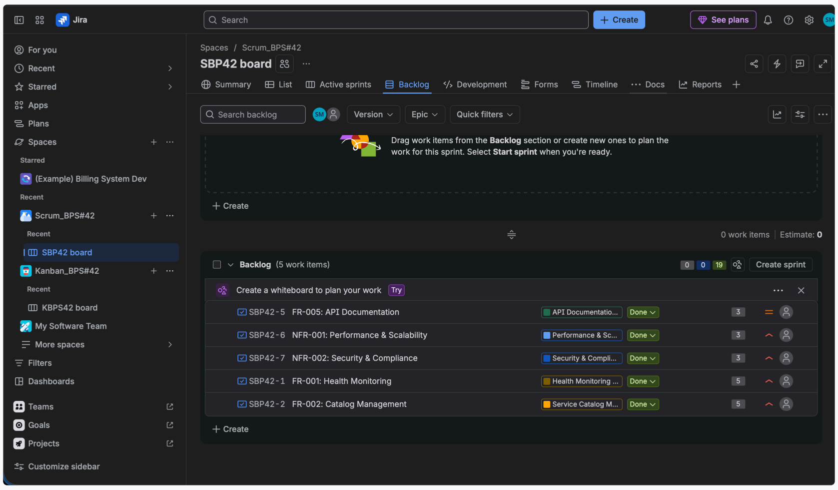
Description: Started Sprint 1 and simulated agile task progression on the Active Sprints board. Advanced FR-001 and FR-002 through *IN PROGRESS* and completed FR-003 and FR-004 to *DONE*.



4. Completed Sprint & Backlog Status

Stories Completed & Accepted

Description: Product backlog state reflecting completed sprints, with user stories transitioning to *Done* status and story points contributing to cumulative team velocity.



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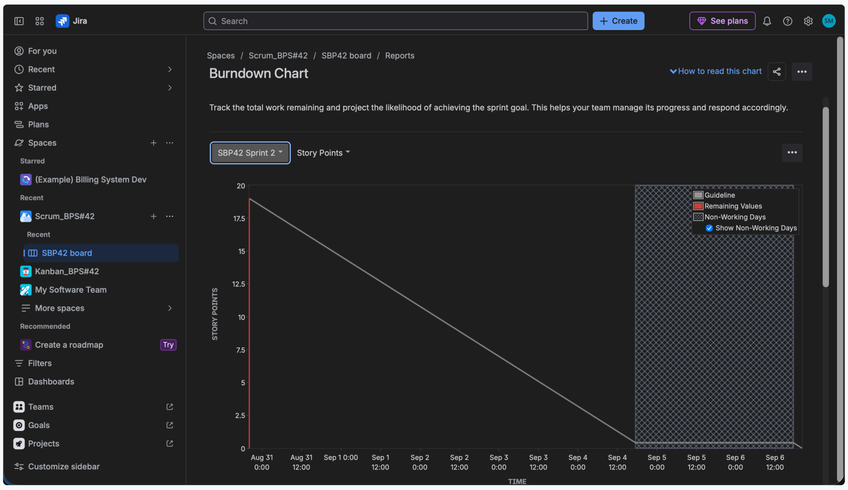
5. Jira Burndown Chart

Sprint Burndown: Remaining Story Points vs. Time

Description: Official Jira Burndown Chart illustrating story point completion across the sprint timeline. Demonstrates work tracking from initial estimation (~19 SP) down to zero remaining work against the ideal guideline.



6. Agile Reflection & Analysis

Lab 2 Post-Sprint Evaluation

Q1: Did your estimations reflect the actual effort?
Yes, using the Fibonacci scale (1, 2, 3, 5, 8) accurately differentiated high-complexity architecture stories (5 pts for health pinger engine, catalog management, and dependency graph) from lower-complexity documentation and security enforcement tasks (3 pts).

Q2: Was your backlog well-prioritized?
Yes, core functional foundations (FR-001 Health Monitoring and FR-002 Catalog) were prioritized as High and scheduled in Sprint 1, delivering an MVP before moving to documentation and performance optimizations in Sprint 2.

Q3: How did your simulated sprint align with your plan?
The sprint execution followed the planned 1-week timebox with linear progression from To Do → In Progress → Done, maintaining zero scope creep and achieving 100% completion of committed story points.

Q4: What insights did the burndown chart give about your team's capacity?
The burndown chart confirmed steady velocity. The remaining work curve tracked closely with the linear grey guideline, indicating healthy sprint pacing without last-minute panic or bottlenecks.

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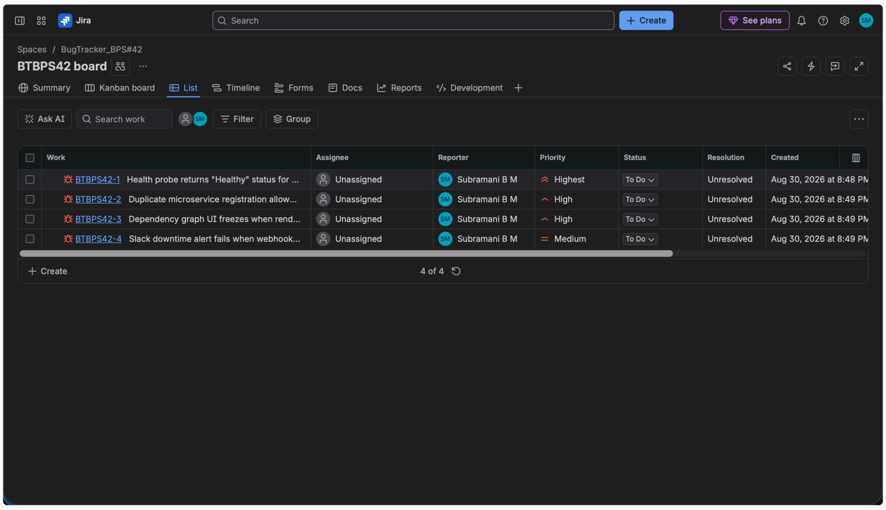
Lab 2: Jira Bug Tracking Project Report

Course: Software Engineering Lab 2 • Problem Statement #42: Internal Microservice Catalog & Health Portal

1. Bug Tracker Space & Defect Backlog View

Project: BugTracker_BPS#42 | Key: BTBPS42

Description: Created a dedicated Bug Tracking workspace in Jira named `BugTracker_BPS#42` with issue key prefix BTBPS42. Reported defects across core functional modules including health monitoring, catalog registration, dependency visualization, and alert dispatchers.



Defect Inventory Summary

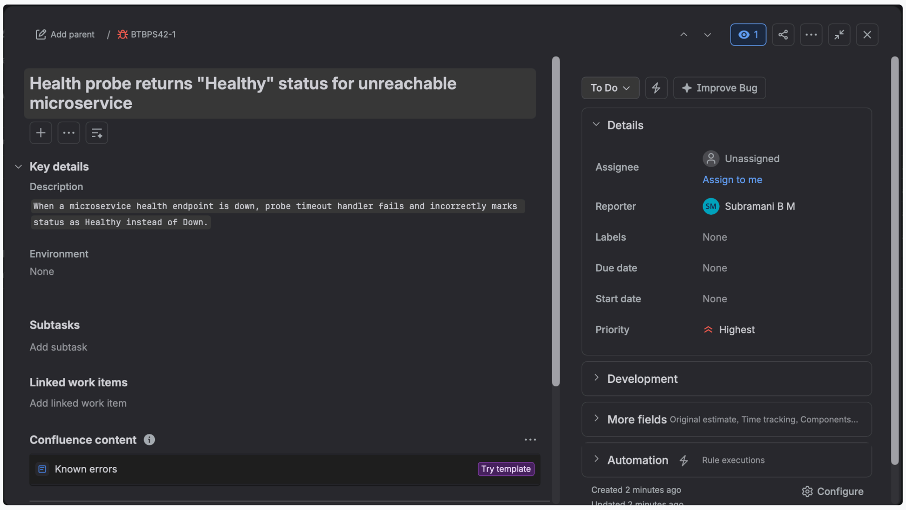
Reporter: Subramani B M

Issue Key	Bug Summary	Component	Priority	Status
BTBPS42-1	Health probe returns "Healthy" status for unreachable microservice	Health Monitoring (FR-001)	Highest	To Do
BTBPS42-2	Duplicate microservice registration allowed with identical names and URLs	Catalog Mgmt (FR-002)	High	To Do
BTBPS42-3	Dependency graph UI freezes when rendering 150+ service nodes	Performance (NFR-001)	High	To Do
BTBPS42-4	Slack downtime alert fails when webhook URL contains encoded characters	Alerting (FR-004)	Medium	To Do

2. Detailed Defect Inspection View (BTBPS42-1)

Priority: Highest • Severity: Critical

Description: In-depth defect record for BTBPS42–1 documenting failure of the probe timeout handler to propagate connection refusal errors, resulting in false-positive health statuses for downed microservices.



3. Defect Lifecycle & Resolution Strategy

Engineering Action Plan

Root Cause Analysis: The HTTP probe pinger client did not catch `ECONNREFUSED` socket exceptions correctly, causing the status evaluator to treat uncaught exceptions as inconclusive rather than unreachable.

Resolution Plan: Wrap health probe network calls in an explicit timeout guard with strict status code checks (200-299 = Healthy, any timeout / socket error = Down). Fix will be validated through automated chaos testing in Sprint 2.